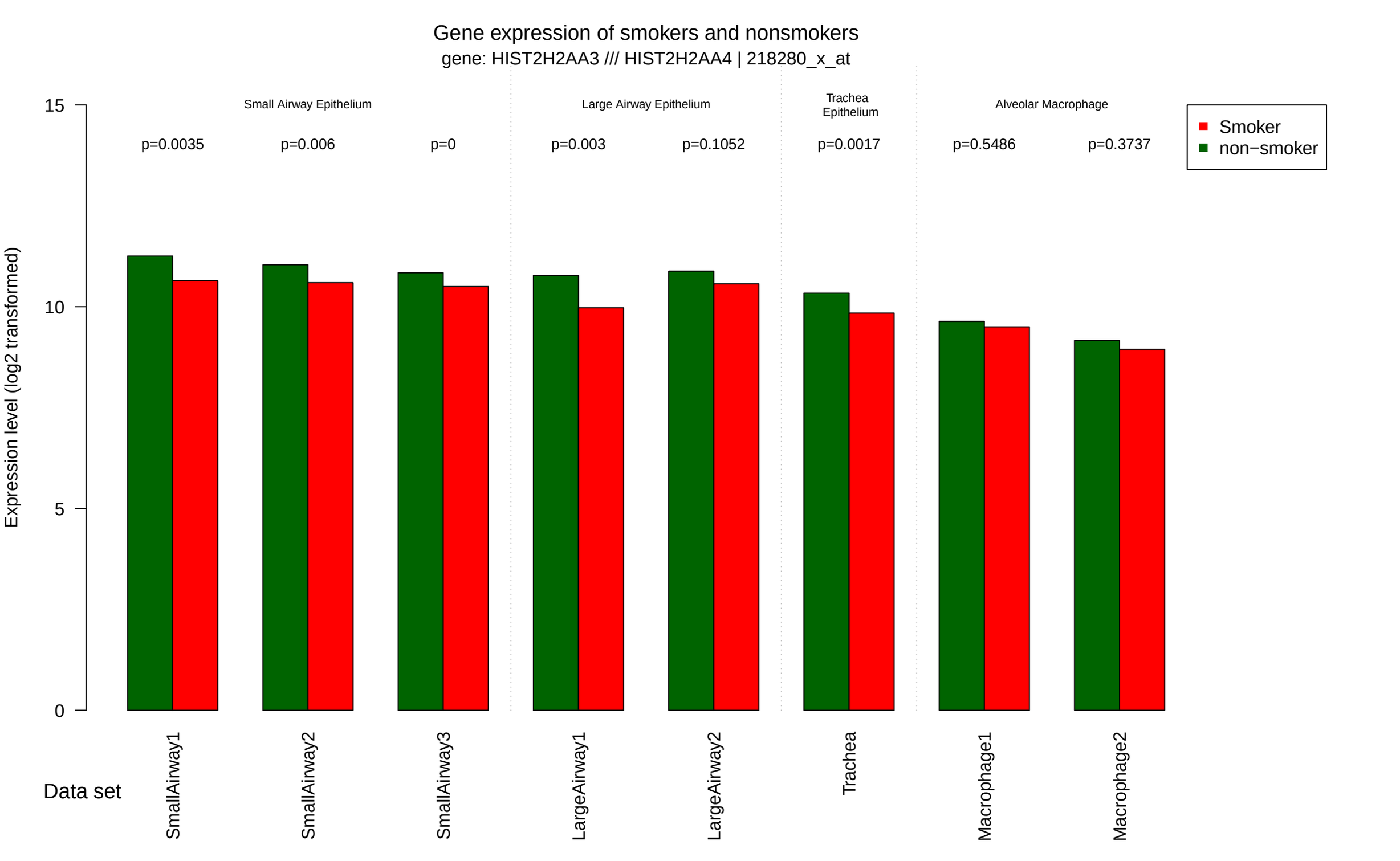
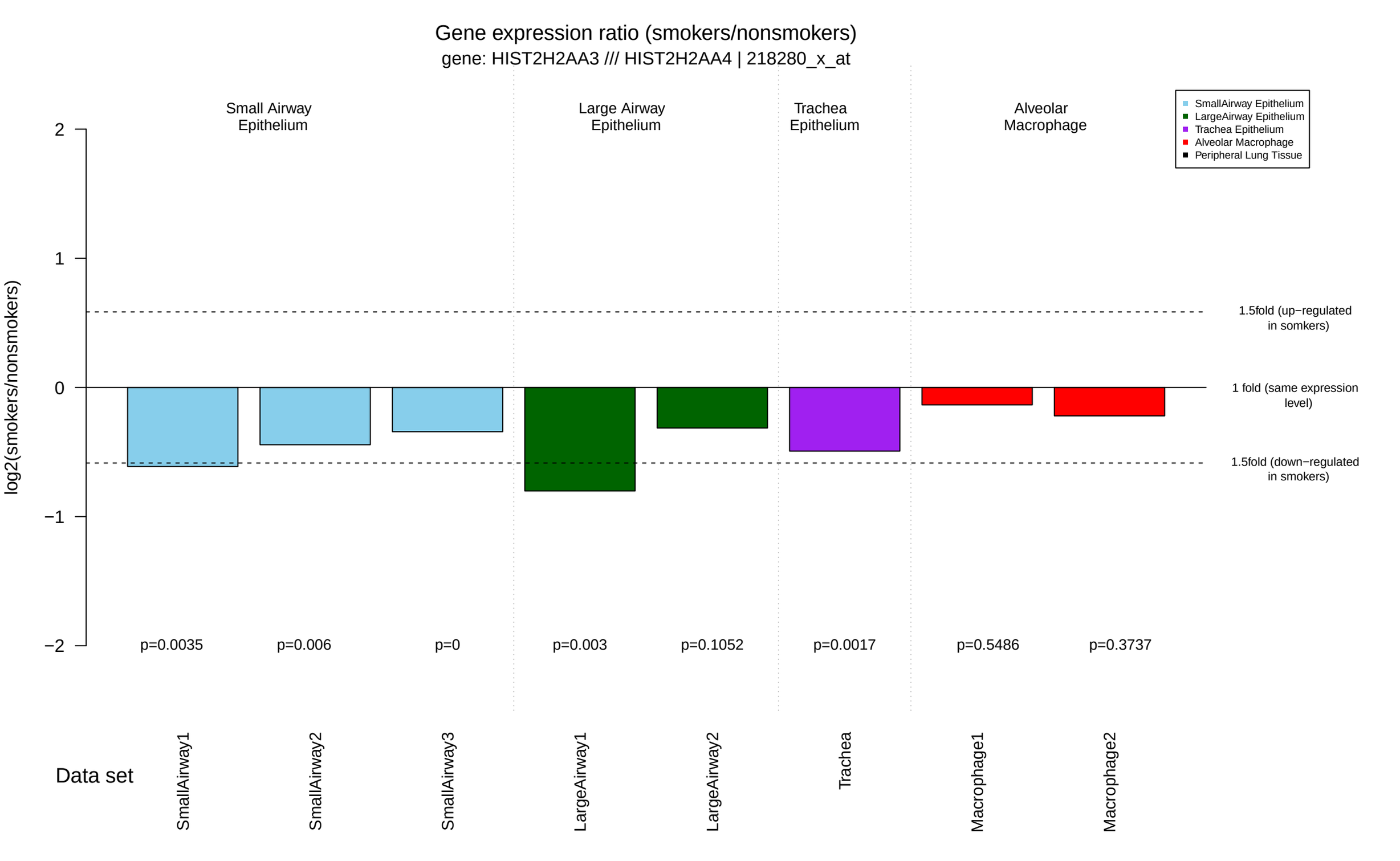
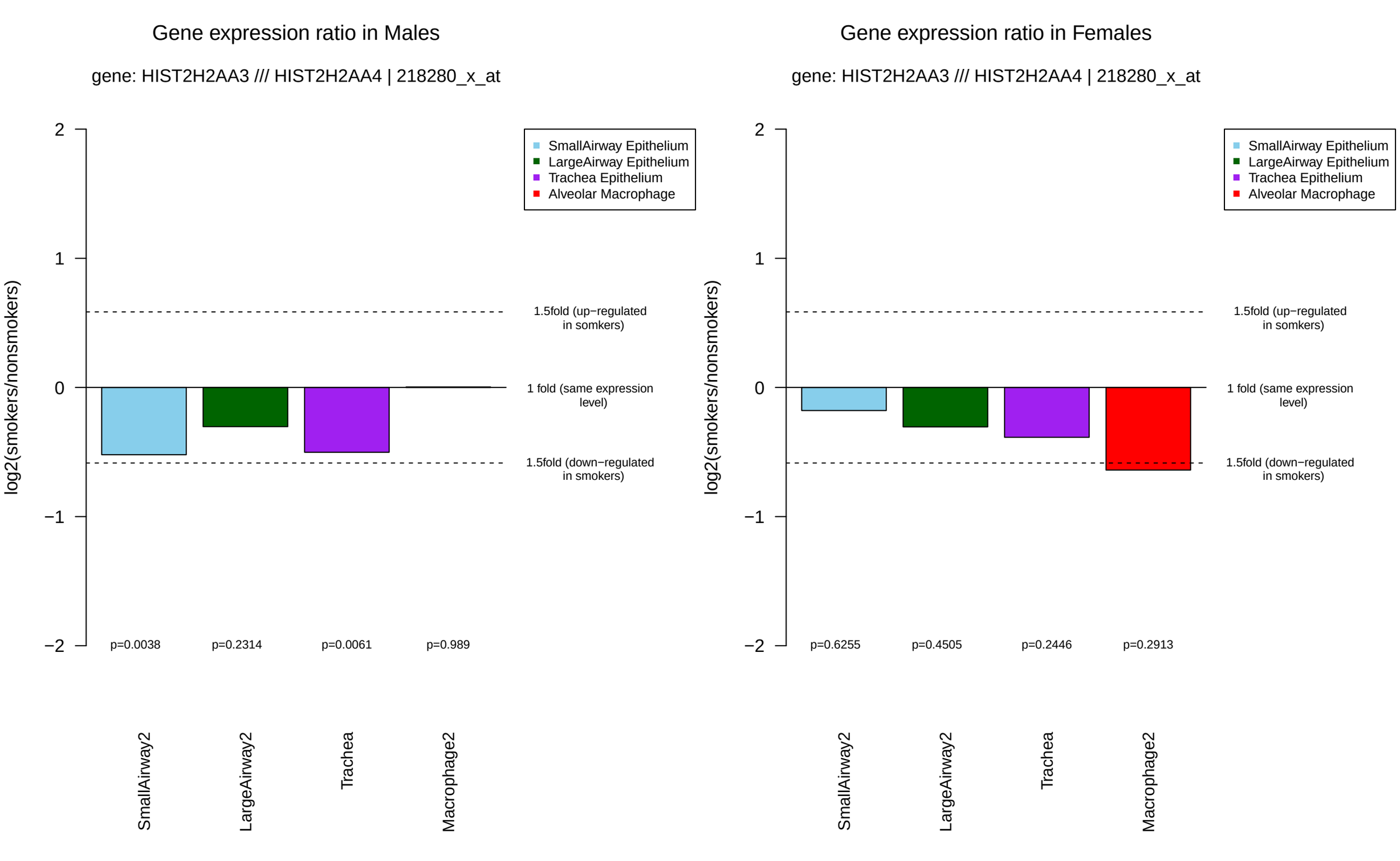
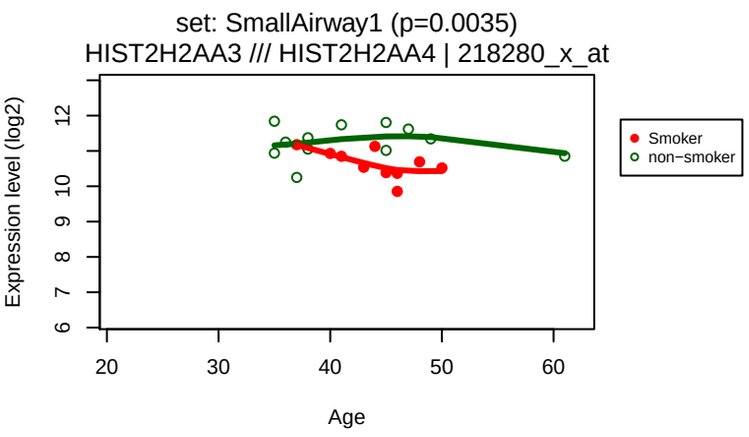
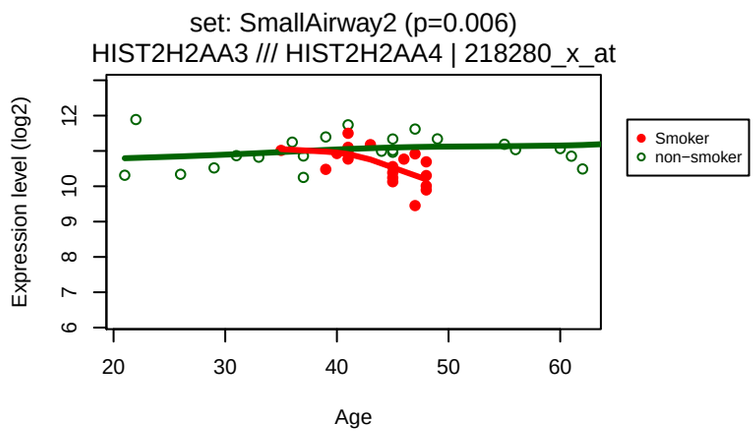
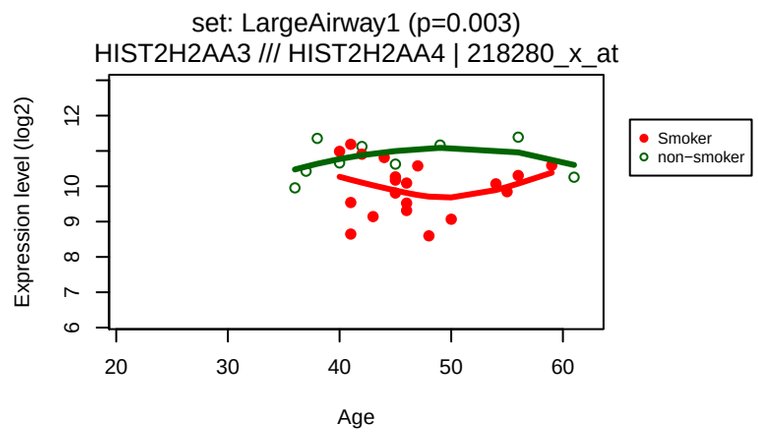

Gene expression of smokers and nonsmokers
gene: HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0035
p=0.006
p=0
p=0.003
p=0.1052
p=0.0017
p=0.5486
p=0.3737
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

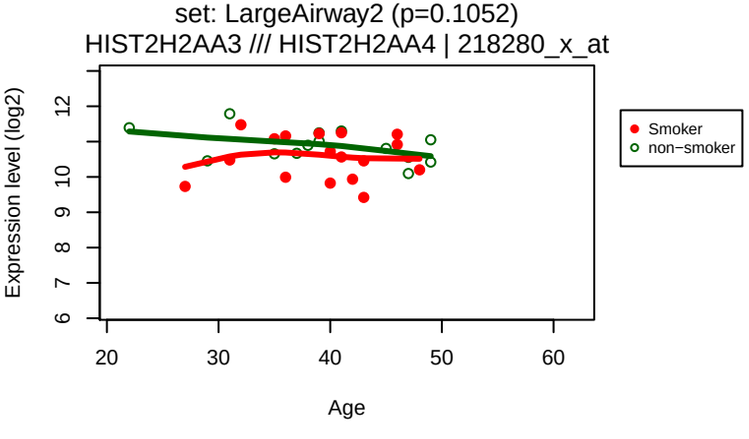

Gene expression ratio (smokers/nonsmokers)
gene: HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0035
p=0.006
p=0
p=0.003
p=0.1052
p=0.0017
p=0.5486
p=0.3737
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

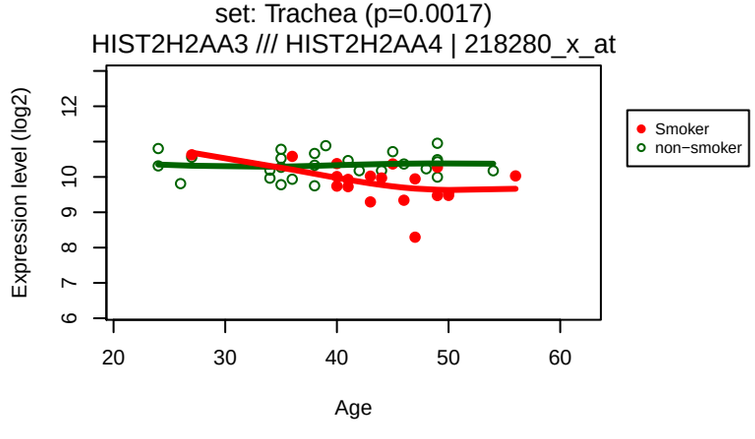

Gene expression ratio in Males
gene: HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0038
p=0.2314
p=0.0061
p=0.989
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6255
p=0.4505
p=0.2446
p=0.2913
SmallAirway2
LargeAirway2
Trachea
Macrophage2

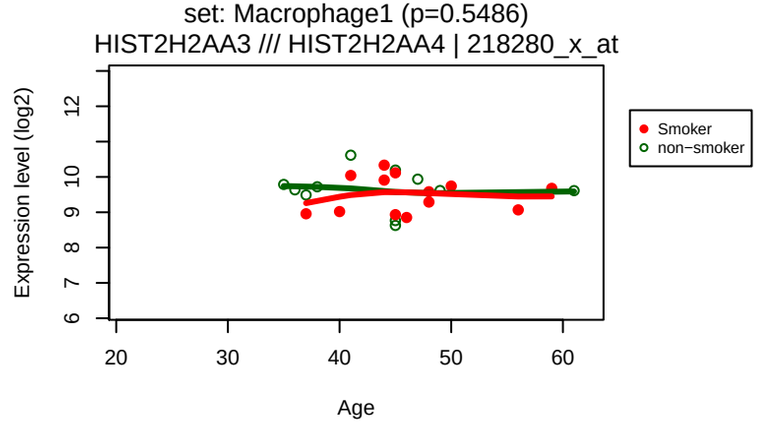

set: SmallAirway1 (p=0.0035)
HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Expression level (log2)
Age
Smoker
non−smoker

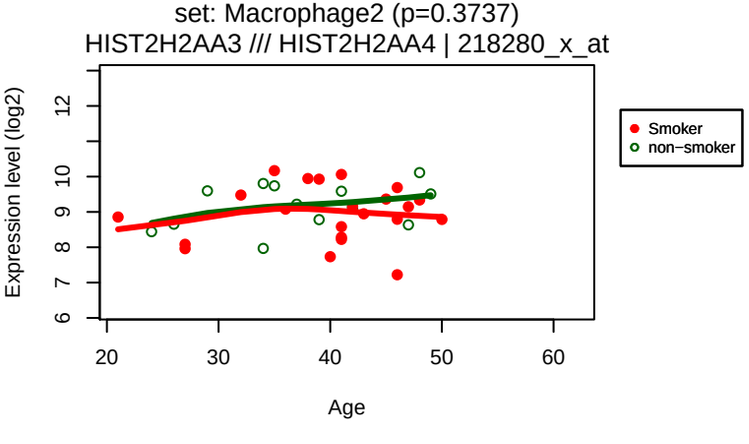

set: SmallAirway2 (p=0.006)
HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Expression level (log2)
Age
Smoker
non−smoker

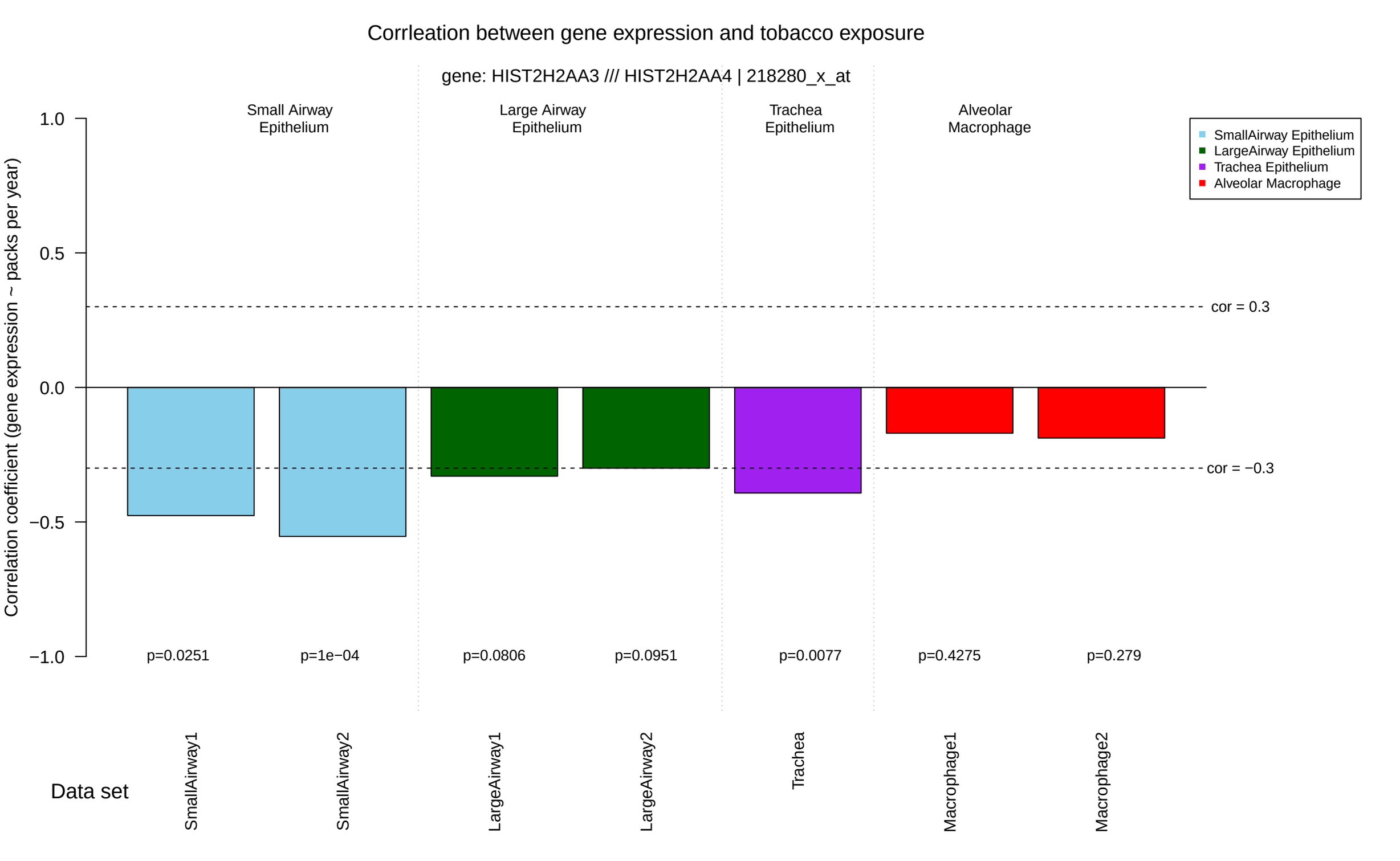

set: LargeAirway1 (p=0.003)
HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1052)
HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0017)
HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5486)
HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3737)
HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIST2H2AA3 /// HIST2H2AA4 | 218280_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0251
p=1e−04
p=0.0806
p=0.0951
p=0.0077
p=0.4275
p=0.279
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2